BDSS Annual Datathon 2026

Evaluating Model Performance vs Environmental Footprint

Credit Card Fraud Detection | Lloyds Banking Group

Can we detect fraud effectively while minimising our carbon footprint?

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The Challenge: Credit Card Fraud Detection

Credit card fraud costs UK consumers and banks billions annually. Financial institutions must detect fraudulent transactions in real time while minimising false positives that disrupt legitimate customers.

The challenge: newer, more complex models promise better detection — but at what environmental cost?

Our mission: compare model performance against carbon footprint and find the optimal balance.

Metric	Value
Total Transactions	284,807
Fraud Cases	492
Fraud Rate	0.173%
Features	30
PCA Components	V1-V28 (anonymised)
Train/Test Split	80% / 20%

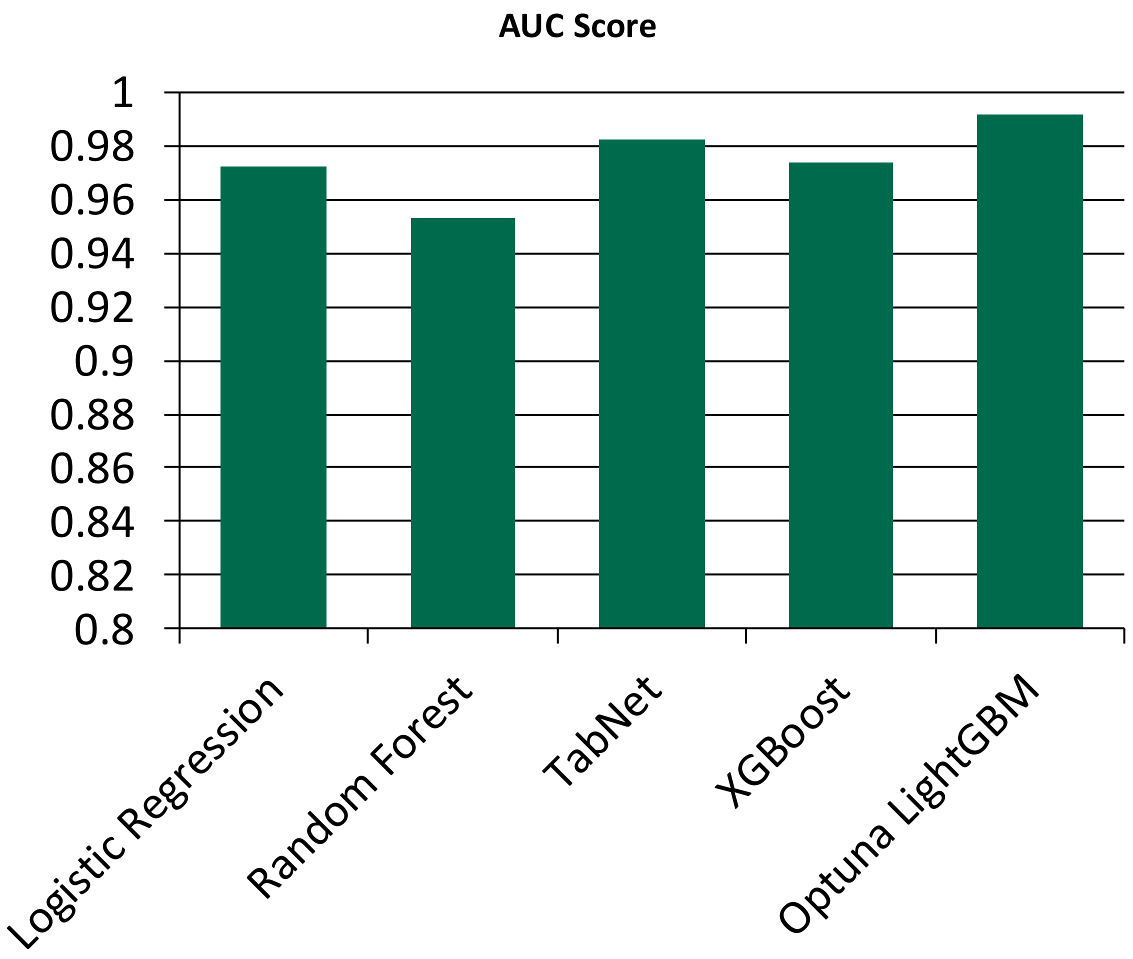
Classification Performance (Threshold-Tuned)

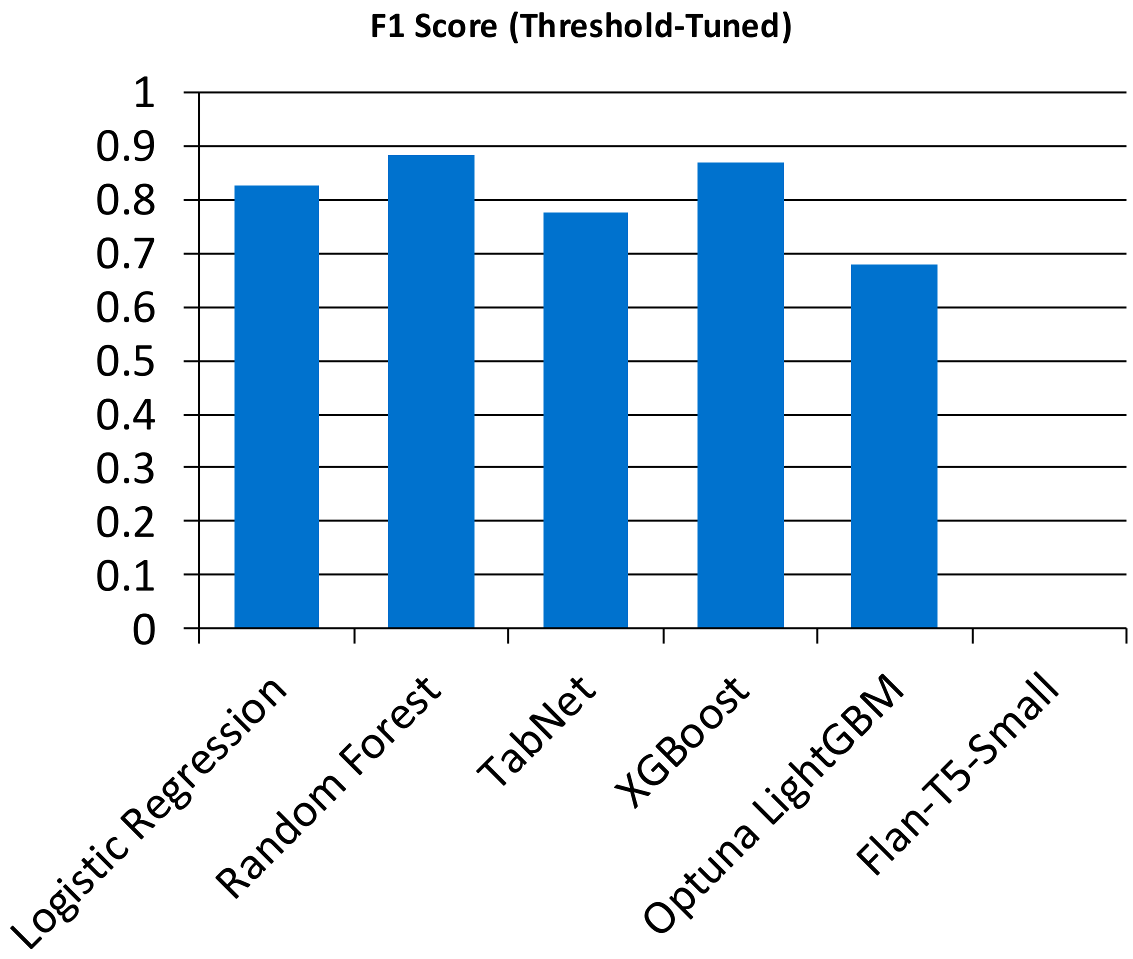
Model	AUC	F1	Precision	Recall
Logistic Regression	0.9722	0.8247	0.8333	0.8163
Random Forest	0.9530	0.8817	0.9318	0.8367
TabNet	0.9824	0.7745	0.7453	0.8061
XGBoost	0.9737	0.8678	0.8958	0.8367
Optuna LightGBM	0.9916	0.6781	0.5852	0.8061
Flan-T5-Small	N/A	0.0000	0.0000	0.0000

Key Observations:

- XGBoost achieves the best F1 (0.87) with excellent precision (0.90) and low carbon
- Random Forest is close behind (F1=0.88) — both gradient boosting models dominate
- TabNet has the highest AUC (0.98) but lower F1 — AUC alone is misleading for imbalanced data
- Flan-T5 completely fails (F1=0.00) — generative models are wrong for tabular classification

Performance Comparison



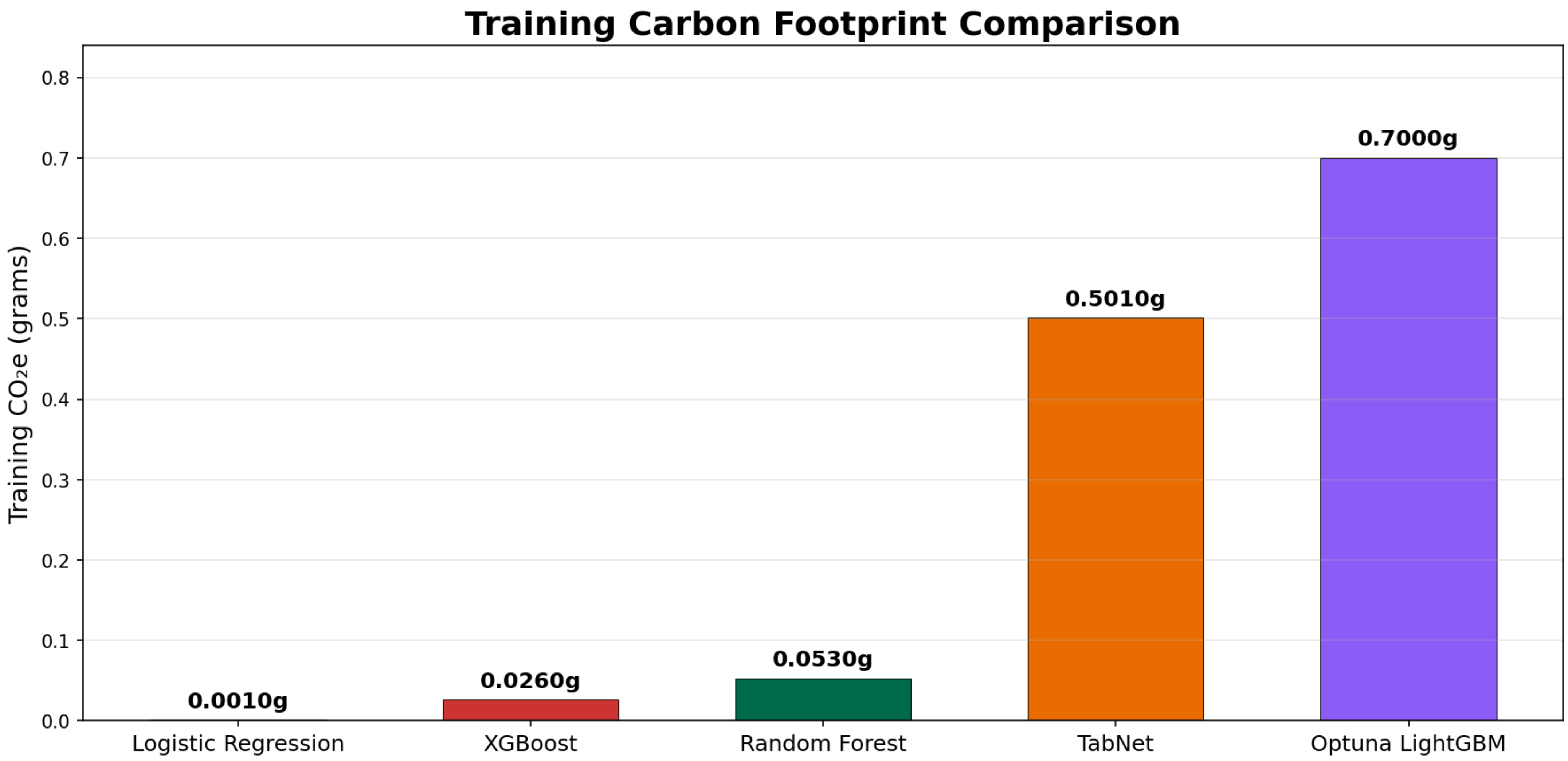


Environmental Impact

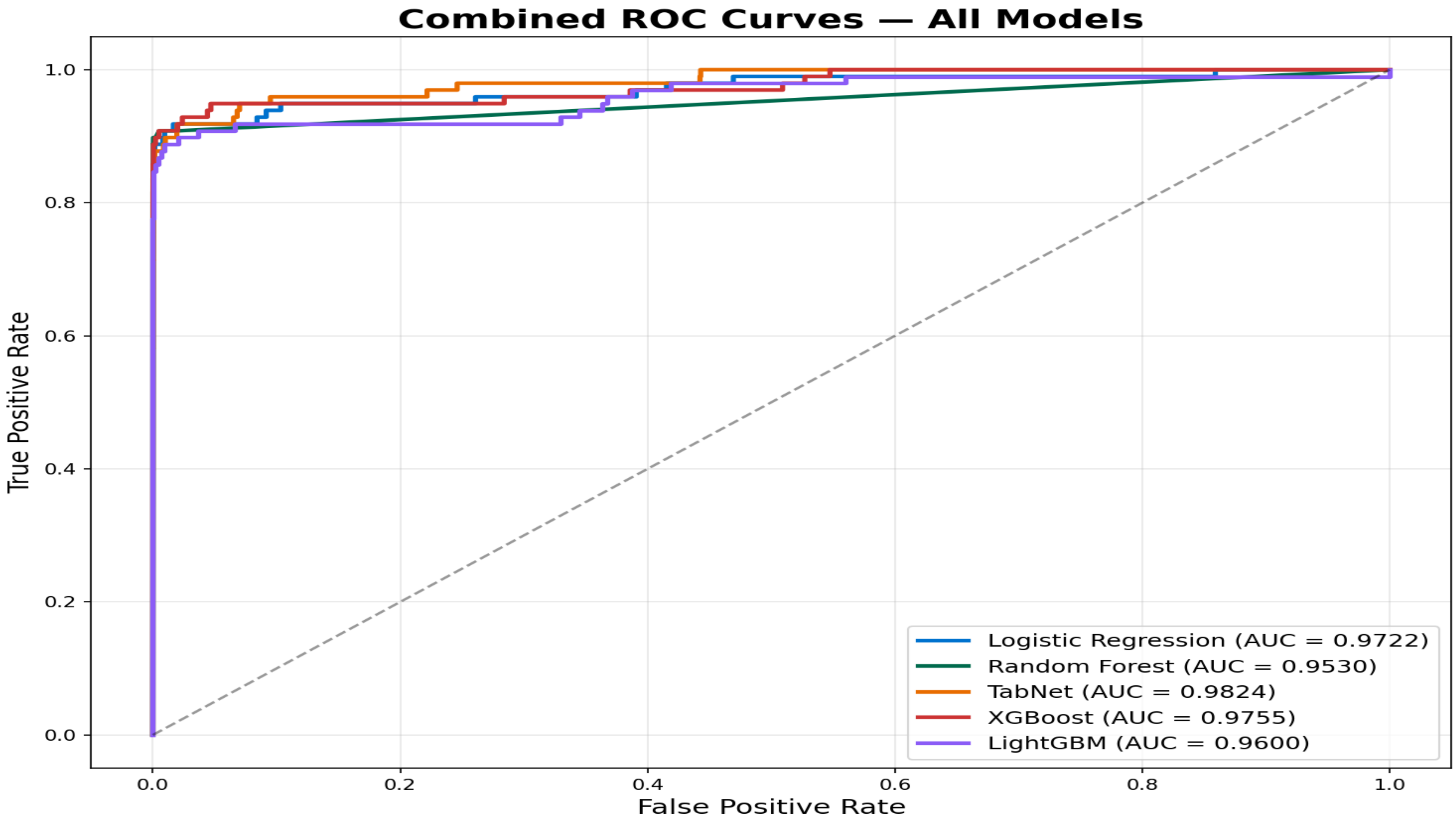
Model	Train Time (s)	Train kWh	Train CO_2e (kg)	Inference CO_2e per 1k (g)
Logistic Regression	0.2	0.000003	0.000001	0.000000
Random Forest	17.8	0.000225	0.000053	0.000002
TabNet	166.7	0.002107	0.000501	0.000031
XGBoost	13.0	0.000110	0.000026	0.000000
Optuna LightGBM	510.5	0.000196	0.000700	0.000001
Flan-T5-Small	1429.8	0.018071	0.004293	0.267646

Key Observations:
- XGBoost: best F1 with only 0.026g CO_2 — trains in 13 seconds
- Logistic Regression: near-zero carbon (0.001g) — trains in 0.2s
- LightGBM Optuna: 50 tuning trials cost 0.7g — 27x more than a single run
- Flan-T5: 4.29g CO_2 for zero fraud detected — worst efficiency by far

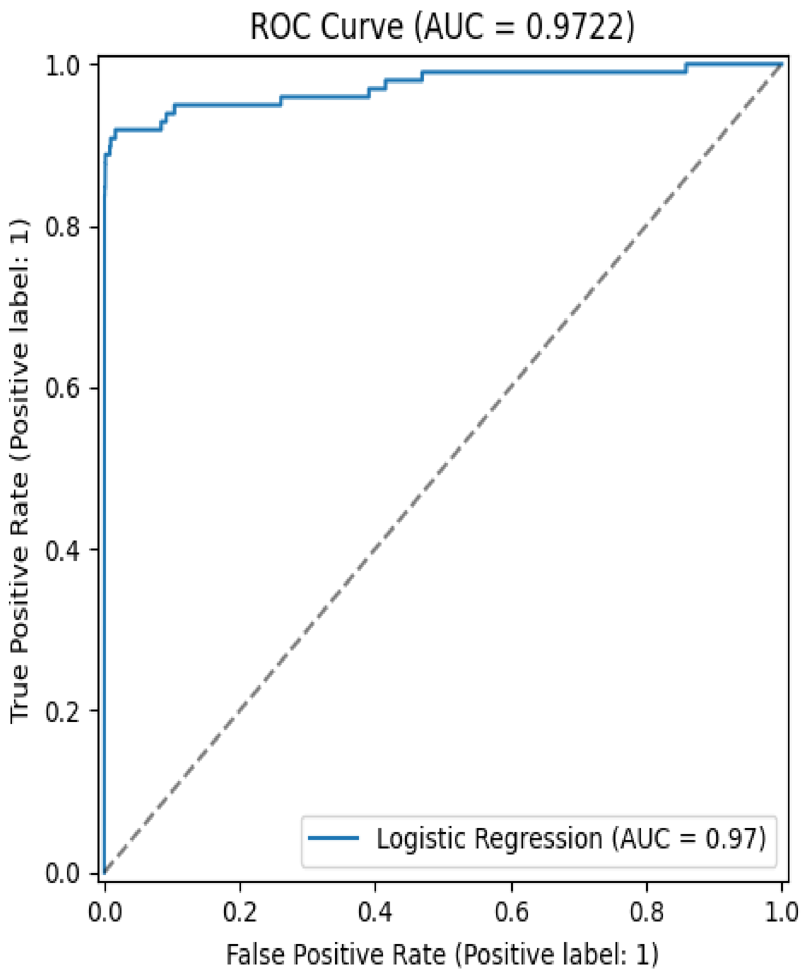
Training Carbon Footprint Comparison



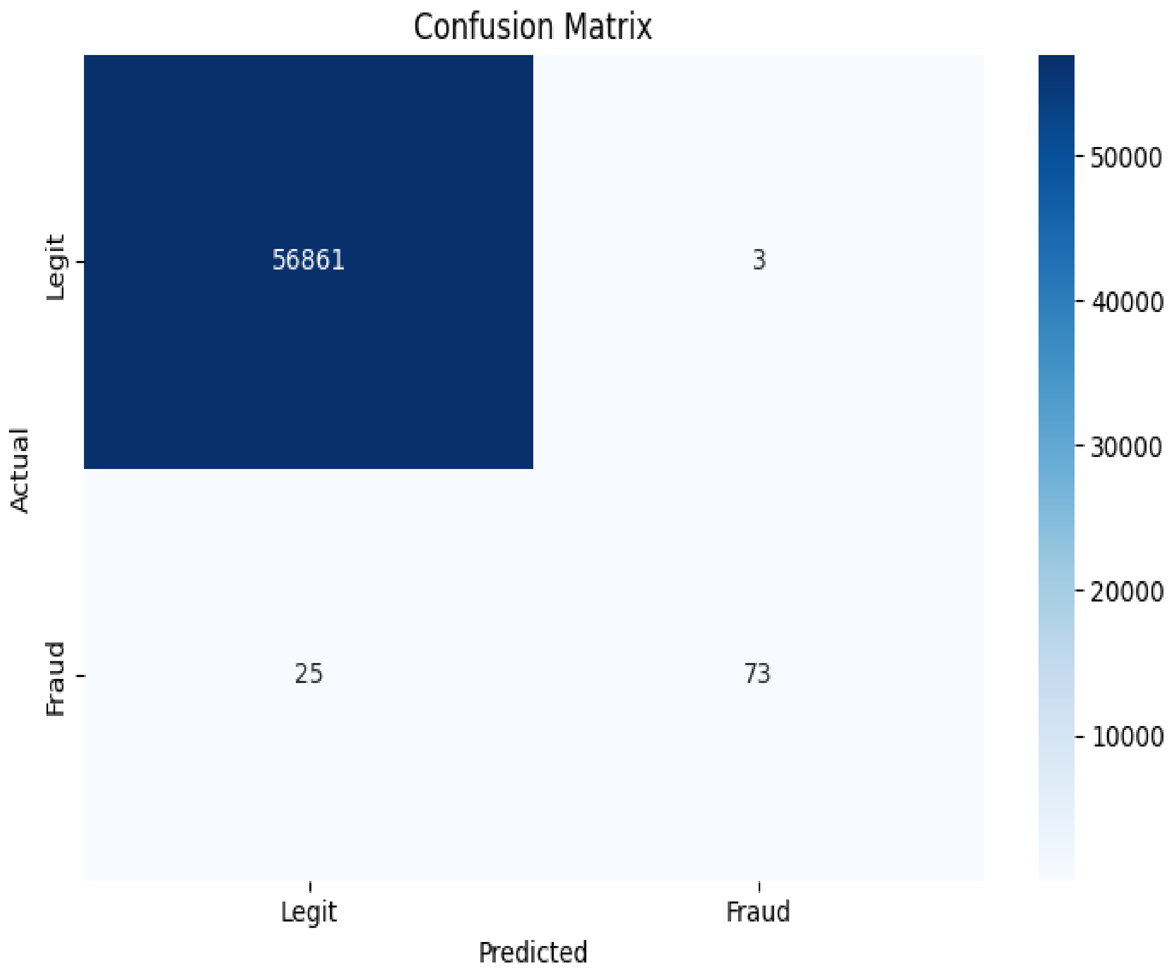
Combined ROC Curves — All Models

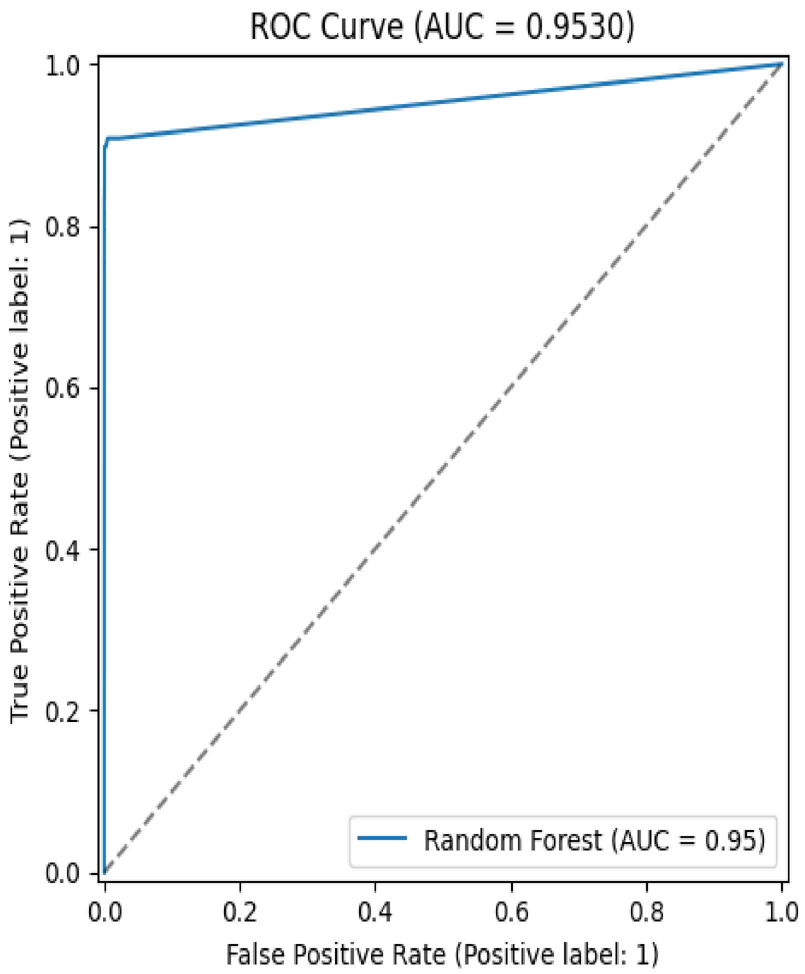


Logistic Regression — Confusion Matrix & ROC Curve

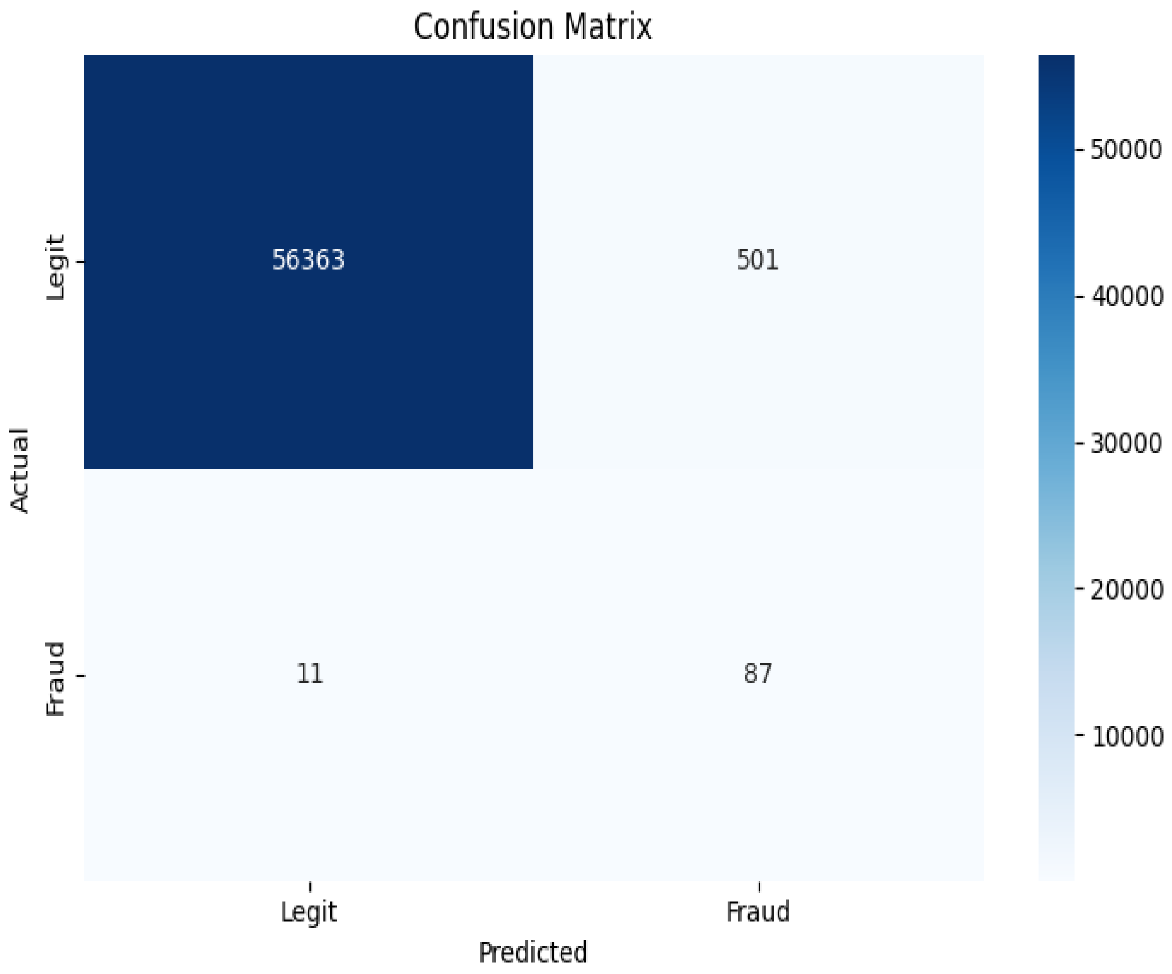


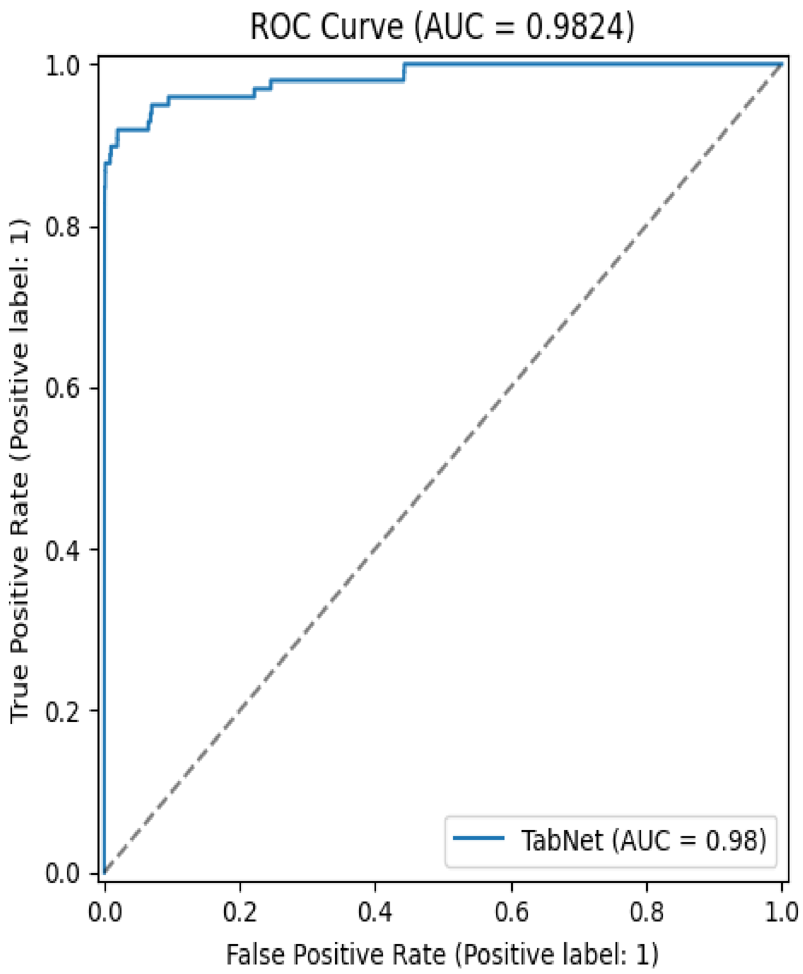
Random Forest — Confusion Matrix & ROC Curve



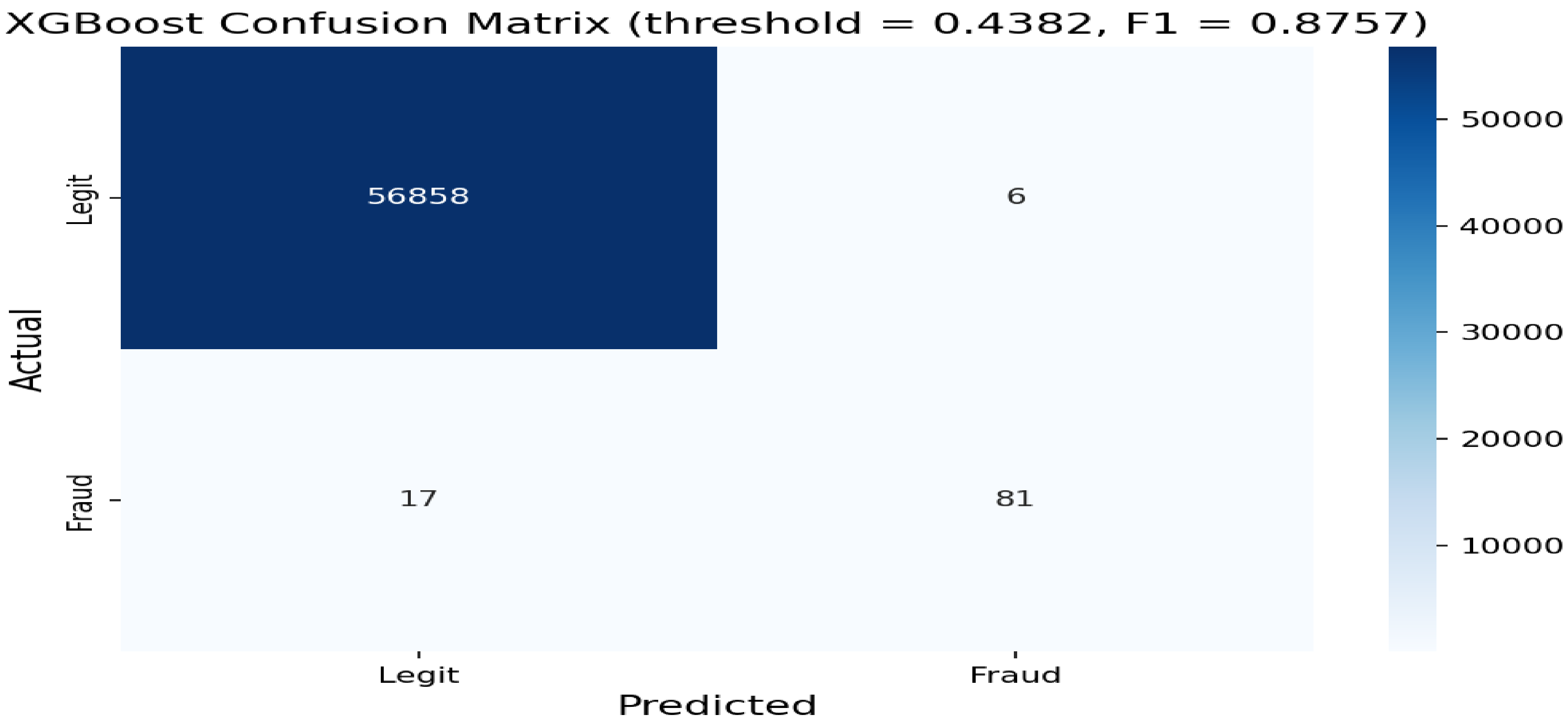


TabNet — Confusion Matrix & ROC Curve

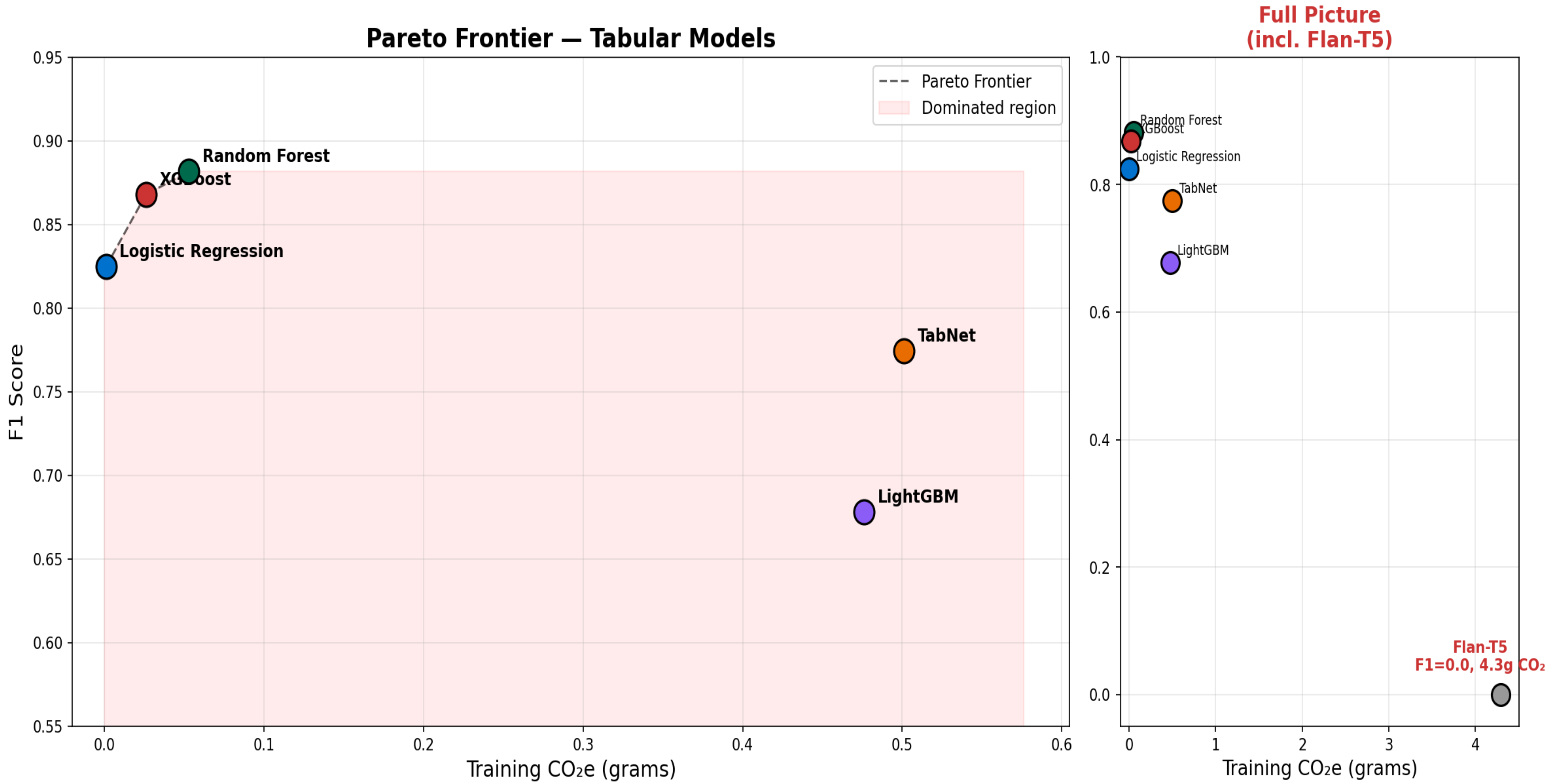




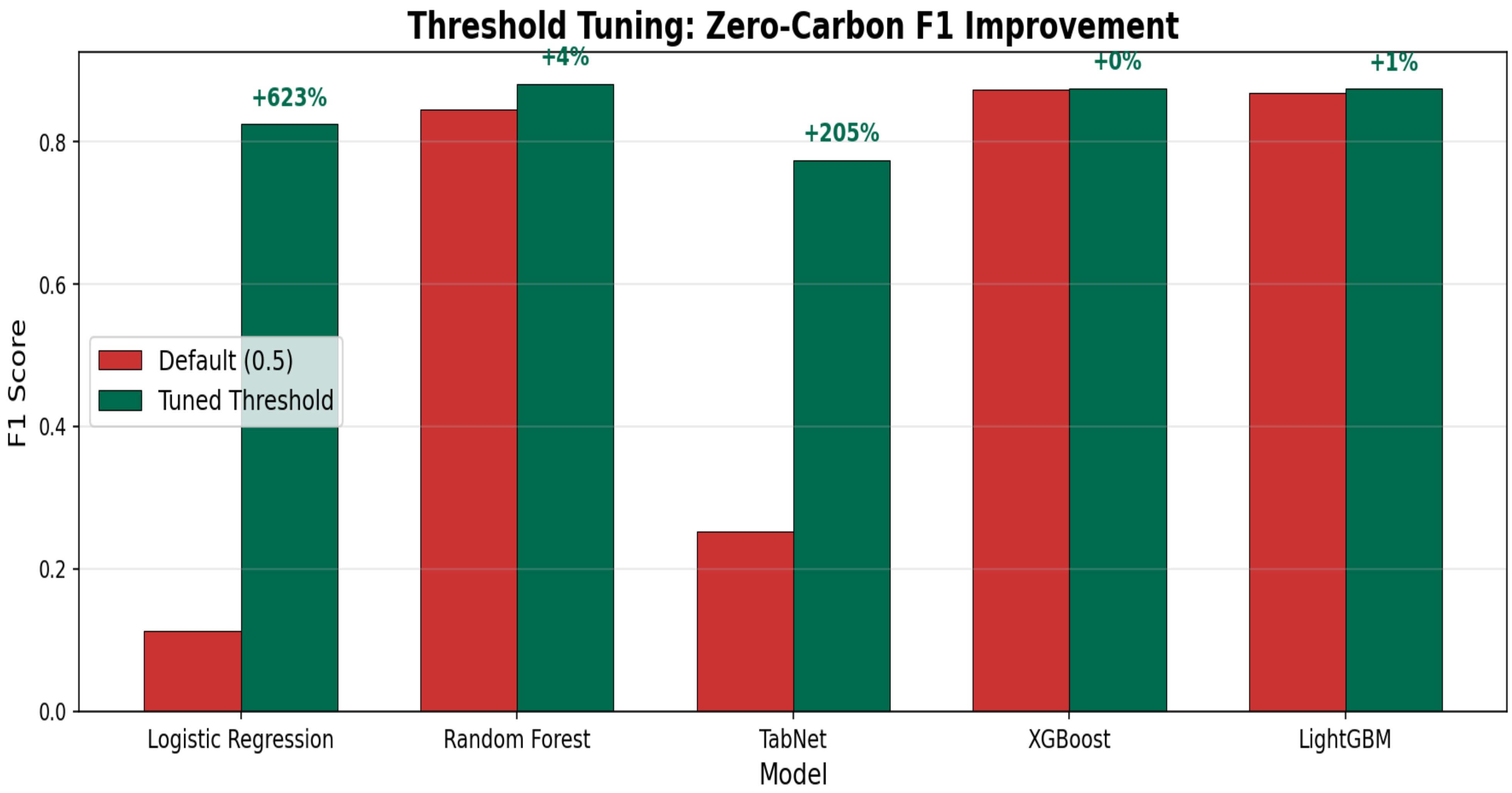
XGBoost — Confusion Matrix



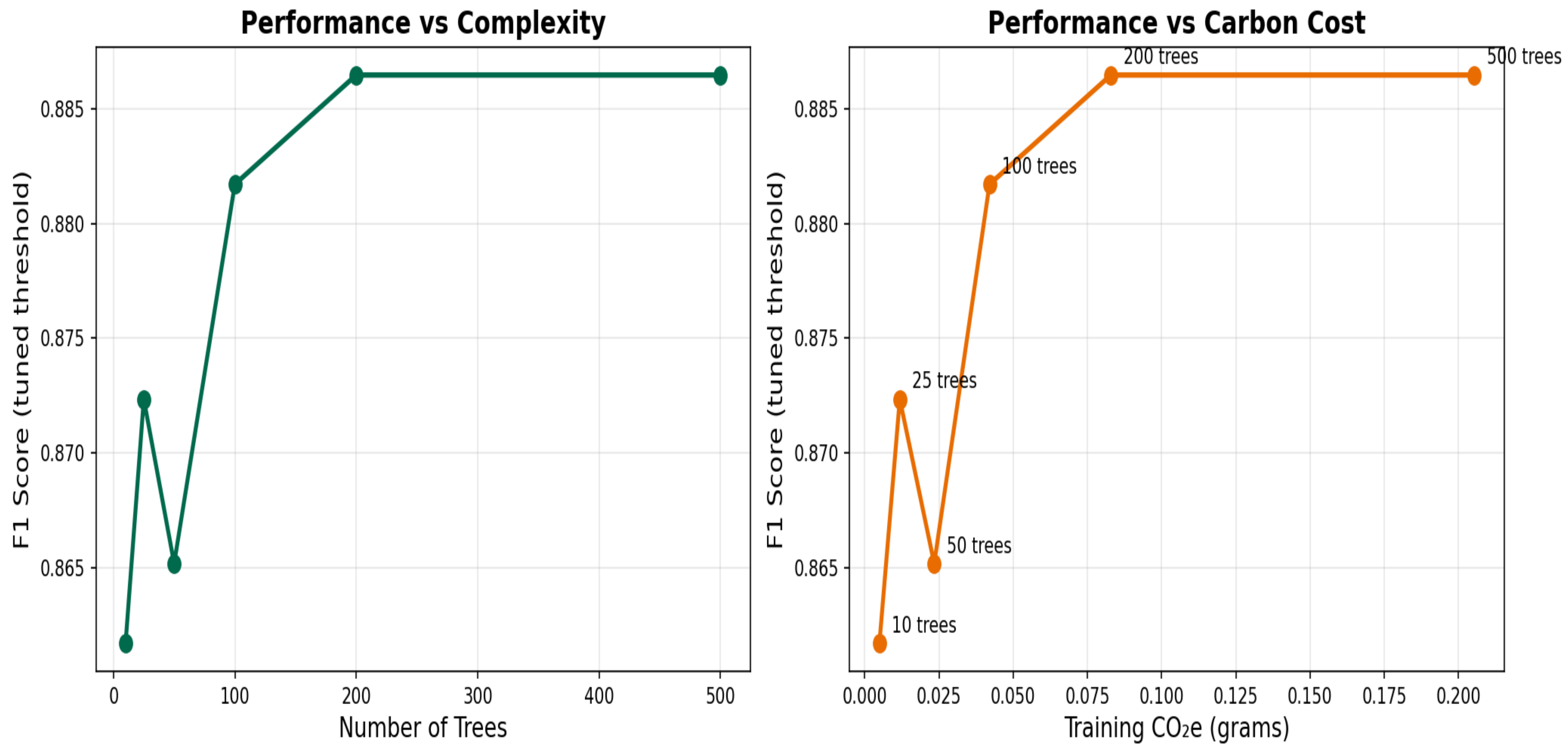
Pareto Frontier — Performance vs Carbon Cost



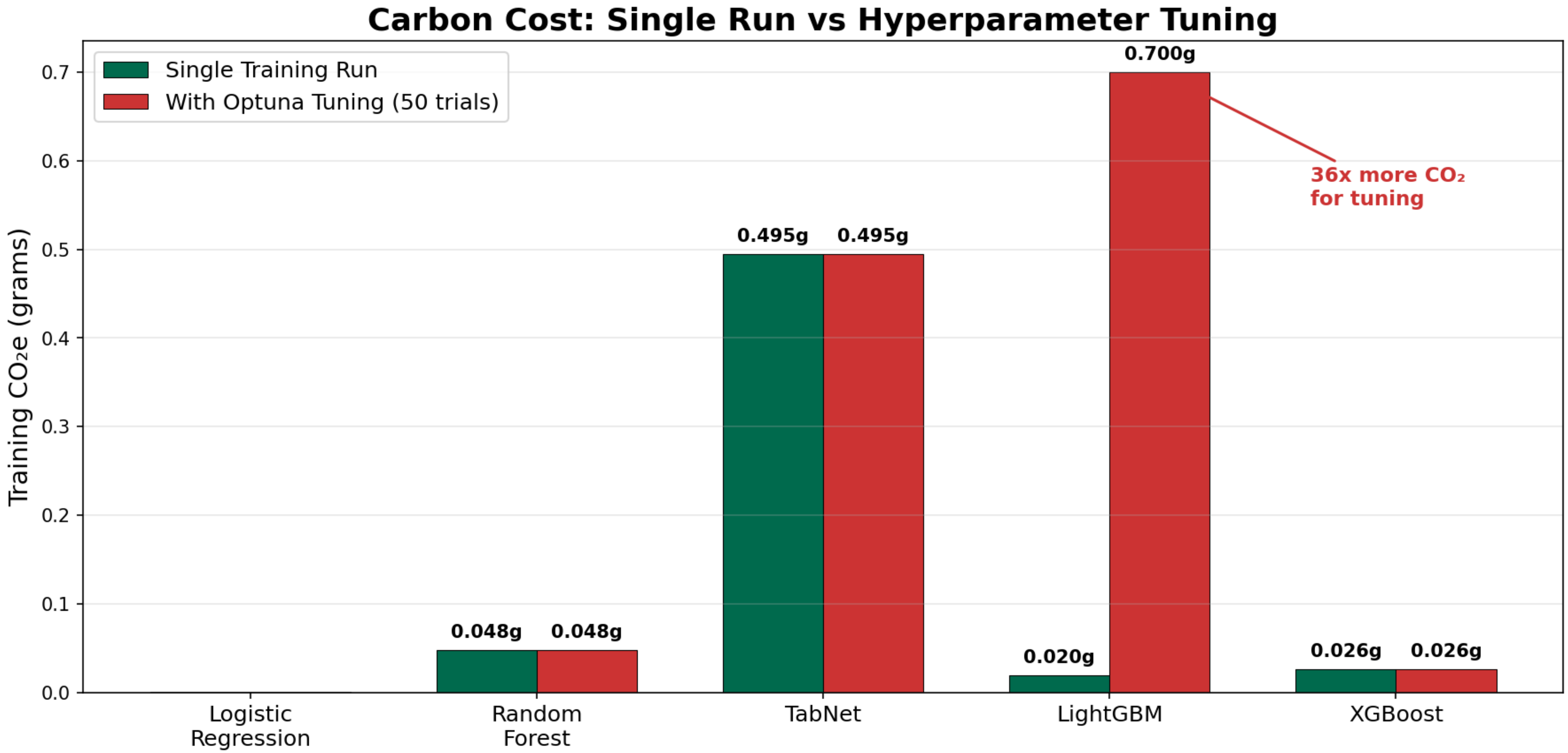
Threshold Tuning — Zero-Carbon F1 Improvement



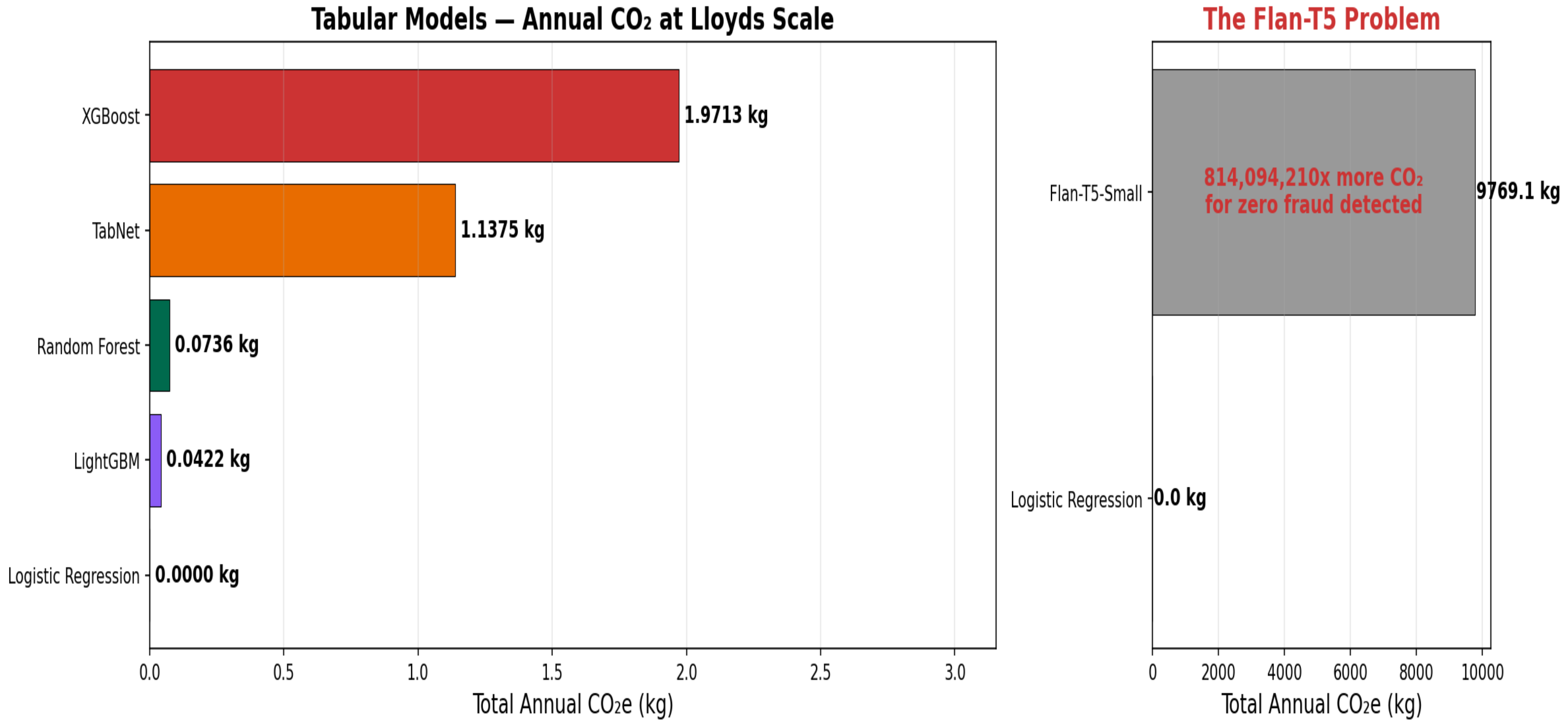
Diminishing Returns — Random Forest (Trees vs Carbon)



Carbon Cost: Single Run vs Hyperparameter Tuning



Annual CO_2 Projection at Lloyds Scale (100M txn/day)



Flan-T5: Why Generative Models Fail on Tabular Data

Results: Complete Failure

Metric	Value
F1 Score	0.0000
Precision	0.0000
Recall	0.0000
Predictions	"legit" for ALL 2,000 samples
Fraud detected	0 out of 3
Train Time	1,430 seconds (24 min)
Train CO_2e	4.29g (4,300x more than LogReg)
Inference CO_2e / 1k	0.268g (10,000x more than RF)

Why It Fails

1. Wrong tool — T5 is for NLP text, not tabular numbers

2. Tokeniser destroys numerical meaning — loses mathematical relationships

3. 60M parameters for what LogReg does with 30 coefficients in 0.2s

4. No probability output — threshold tuning impossible

5. Class imbalance — learns to always predict 'legit'

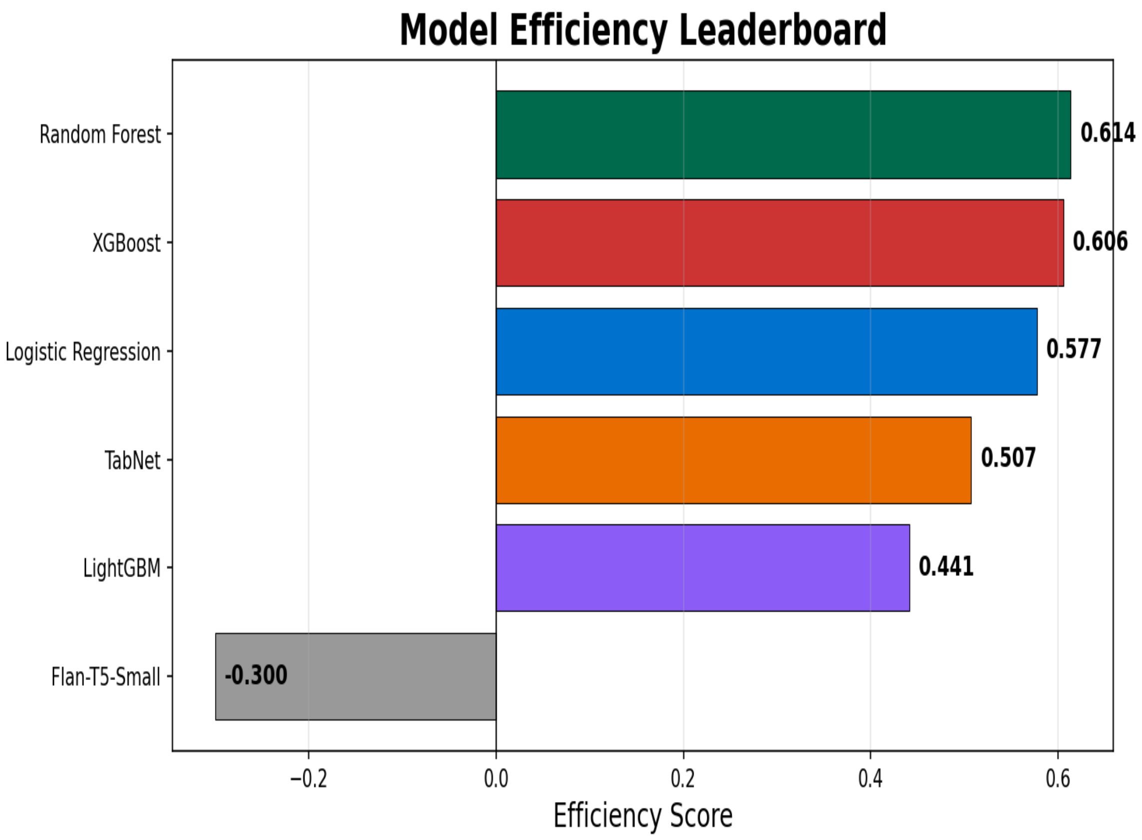
Generative models are fundamentally unsuited for structured tabular classification. Flan-T5 burned 4,300x more carbon than Logistic Regression and detected zero fraud.

Efficiency Leaderboard

Efficiency Score = 0.7 x F1 - 0.3 x Normalised CO_2e

F1 is already 0-1. CO_2 is normalised: (CO_2 - min) / (max - min). 70/30 split: performance matters more, but carbon is penalised.

Rank	Model	F1	CO_2e (g)	Norm CO_2	Efficiency Score
1	Random Forest	0.8817	0.0530	0.0121	0.6136
2	XGBoost	0.8678	0.0260	0.0058	0.6057
3	Logistic Regression	0.8247	0.0010	0.0000	0.5773
4	TabNet	0.7745	0.5010	0.1165	0.5072
5	Optuna LightGBM	0.6781	0.7000	0.1629	0.4258
6	Flan-T5-Small	0.0000	4.2930	1.0000	-0.3000



Key Insights & Recommendations

XGBoost wins the efficiency leaderboard

Best F1 (0.87) with only 0.026g CO_2 — trains in 13 seconds. Random Forest (F1=0.88) is a close second with 0.048g. Both gradient boosting models dominate.

Threshold tuning is the best free optimisation

Tuning the classification threshold improved TabNet F1 from 0.25 to 0.77 and LogReg from 0.11 to 0.82 with zero extra carbon. Every model should do this before deployment.

Hyperparameter tuning has a hidden carbon cost

LightGBM's 50 Optuna trials cost 0.7g CO_2 — 36x more than a single run. The performance gain must justify the search cost.

Generative models are wrong for tabular data

Flan-T5 burned 4,300x more carbon than LogReg and detected zero fraud. NLP models destroy numerical meaning through tokenisation.

Recommendation: deploy XGBoost with threshold tuning

Or use a two-stage pipeline: LogReg as a fast filter (near-zero carbon), then XGBoost/RF only on flagged transactions — saving ~90% of inference carbon.

Thank You

Questions?

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